Problem Chosen: A 2025 HiMCM
Team Control Number: 16603

Summary Sheet

A Three-Stage Optimize–then–Simulate Model for Rapid Building Clearance Under Disaster Conditions

Summary

Introduction: The rapid and complete sweep of a building during an emergency is a critical, life-saving operation. However, emergency planners often lack the tools to create robust plans that account for the chaotic, probabilistic nature of disasters. This paper presents a 3-stage "Optimize–then–Simulate" hybrid model designed to find the optimal responder assignment (K) and estimate a realistic, 95th-percentile clearance time (T_{clear}) by modeling real-world physics and random failures.

Methodology: Our model consists of three components: (1) Zone Optimization using K-Means to assign K responders to compact building zones; (2) Path Optimization via a Traveling Salesperson Problem (TSP) heuristic to find the shortest search path within each zone; and (3) Stochastic Simulation (Monte Carlo + A*) to model clearance times with probabilistic disruptions, dynamic rescues, and pathfinding adjustments.

Results: Our "Optimize K" analysis (1,000 runs per K) on a baseline office scenario identified an optimal team size of $K = 3$. The final 100,000-run simulation for $K = 3$ yielded a 95th-percentile T_{clear} of 8.4 minutes, showing that the mean time (4.1 minutes) is a dangerously optimistic metric. When applied to a high-risk School (Fire) scenario, the 95th-percentile T_{clear} increased by roughly 520% to 43.7 minutes, driven by a high probability of multiple, time-consuming A* rescue trips.

Conclusion Recommendations: Our model is a robust tool: the 95th-percentile clearance time, not the mean, is essential for planning. Use $K = 3$ (trade-off of speed/efficiency). Worst-case analysis beats averages.

Keywords: Emergency evacuation, Monte Carlo simulation, K-Means, TSP, A*, stochastic modeling, risk analysis

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1 Introduction and Problem Restatement

1.1 Problem Background and Motivation

During emergencies such as fires or gas leaks, every second is critical to occupant safety. When first responders arrive on scene, they must work against uncertainty: they do not know exactly where individuals remain in the building, which routes are accessible, or where structural or debris blockages may lie. Responders quickly evaluate the most efficient pathway to search every room, continuously recalibrating their paths as new information on obstacles is communicated to them.

These sweeps are difficult because they are strongly affected by uncontrollable factors such as smoke density, collapsed ceilings, jammed doors, and debris-blocked corridors. All of these contribute to the time it takes to reach and save someone in a room. It is therefore essential to model these sweeps mathematically, so that responders and planners can make informed decisions before disasters occur.

1.2 Detailed Problem Restatement and Interpretation

To declare "all clear" for emergency evacuations, several safety and procedural conditions must be satisfied:

- Responders must visually confirm that no persons remain in each room.
- They must perform an audible check, calling out and reporting "clear" on their radios, especially when visibility is poor.
- They must conduct a physical inspection of all subareas in a room (corners, behind furniture, under desks when appropriate).
- They must evaluate residual hazards, including smoke, gas, or structural debris that could still endanger safety.

Only after confirming both human and environmental safety can the area be considered truly "all clear." The room is then clearly marked and documented over radio to prevent re-entry and redundant searches.

When responders encounter remaining individuals, those individuals must be assisted or carried to safety and transferred to medical personnel. Natural randomness in disasters is inevitable. A path that appears optimal in advance may suddenly be blocked by debris or smoke. Responders then attempt to navigate around blocked paths, and in worst cases, may be delayed enough that some occupants cannot be reached in time.

We assume all first responders entering the building are equally healthy, fast, and strong, having been rigorously trained under standard procedure. Other personnel remain on the outer perimeter, supporting triage and treatment once victims are brought outside.

The nature of occupants also plays a critical role in sweep dynamics. Key factors include occupant weight, the number of occupants per room, and the fraction of children or elderly. Facilities with children, such as daycares or schools, require responders to search every corner carefully and assist young individuals who may be hiding. To avoid misinterpreting injuries and needs, responders assist all remaining occupants. Each setting demands a tailored

approach that accounts for room layout, hallway distances, occupant behavior, and building density.

The type of emergency additionally influences sweep operations. Fires, earthquakes, and other hazards create distinct conditions that affect both responder and occupant behavior. In fires, heat, flames, and smoke must be avoided, often forcing responders to prioritize accessible rooms first. Earthquakes can cause structural damage, block exits, and create unstable environments, requiring sweeps that account for debris and alternative escape routes. Each hazard modifies occupant evacuation speed, choice of escape routes, and responder tactics. Our model therefore explicitly distinguishes between fire and earthquake scenarios.

1.3 Sweep Strategy

The model follows a *first-pass sweep strategy*, also known as a level sweep or zone sweep.[1] In this approach, responders conduct an initial search from floor to approximately hip height across a defined zone. The first pass focuses on quickly identifying obvious victims, hazards, or immediate threats. By prioritizing speed and broad coverage over extremely detailed inspection, this strategy ensures that critical areas are rapidly cleared and major dangers are not overlooked during the initial sweep. Detailed, slower secondary searches can occur later if needed; our model specifically optimizes this time-critical first pass.

1.4 Model Pipeline Overview

Our model runs in three main stages:

1. **K-Means clustering** divides the floor plan into zones to assign each responder.
2. **TSP routing** determines the shortest possible route within each assigned zone.
3. **Monte Carlo simulation with A* pathfinding** accounts for real-world uncertainty such as blockages, smoke, varying assistance times, and different occupant weights to produce a full distribution of sweep times and to identify the optimal number of responders K.

[1] FIRESCOPE ICS-500 and FEMA US&R guidance describe level and zone sweeps as standard first-pass tactics.

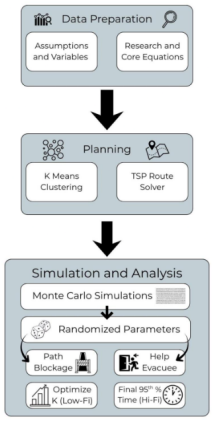


Figure 1: Model pipeline flowchart illustrating K-Means zoning, TSP routing, and Monte Carlo + A* simulation.

2 Preparation and Assumptions

2.1 Modeling Assumptions

To build a realistic yet tractable model for determining "all clear" times, we adopt the following assumptions:

A1: All mobile occupants have already evacuated: Sweeps occur after initial evacuation; only those requiring assistance (children, elderly, injured, or mobility-limited) remain.

A2: All first responders have the same physical capacities: Standardized training means baseline speed, endurance, and strength are uniform.

A3: All remaining occupants require assistance: Anyone inside is unable to evacuate unaided; responders must assist or carry them.

A4: Carrying an occupant reduces responder speed proportionally to occupant weight: Speed decreases with occupant's mass, based on anthropometric data.

A5: Communication among responders is effectively perfect: Radio systems (ER-RCS) are reliable; delays are incorporated as a variable only for blockages.

A6: Visibility penalties apply only to fire scenarios: Only fires generate enough smoke to reduce visibility; earthquakes do not.

A7: No re-entry into cleared rooms: Once searched and marked, rooms are not re-entered, avoiding redundancy per best-practice protocols.

A8: All rescues initiated are successful: If a path exists, responders can transport found occupants; medical mortality and collapse are out of scope.

A9: Maximum of two responders per exit: Limits congestion at exits, following safety protocols.

A10: Responders always use stairs, not elevators: Standard emergency protocol mandates stairs are used only.

A11: Room search is deliberate and slow; fatigue during search is negligible: Search occurs at half normal speed; additional in-room fatigue is insignificant.

A12: Building geometry is accurately known: Floor plans are correct and known pre-event for conversion into graph structures.

2.2 Research Data Supporting the Model

Baseline First Responder Speed

A study of wilderness paramedic candidates carrying approximately 20.4 kg of load on flat terrain reported an average walking speed of 7.6 km/h, or about 2.1 m/s. This value serves as our baseline horizontal speed v_0 for earthquake scenarios (flat indoor surfaces with equipment).For fire scenarios, responders move faster to outrun fire progression and reach victims sooner. We increase v_0 by 0.5 m/s, giving $v_0 = 2.6$ m/s for fires.

Fatigue Constant

Fatigue reduces movement speed over time. In *The effect of fatigue and training status on firefighter performance*, trained firefighters performed a simulated fire-ground test (SFGT) at baseline and after a single fatiguing exercise.[2] The baseline mean completion time was approximately $t_{\text{baseline}} \approx 365.0$ s, and the fatigued completion time increased by 9.6% given by: $t_{\text{fatigued}} \approx 365.0 \times 1.096 \approx 400.0$ s.Assuming a fixed normalized distance $d = 1$ m, the effective speeds would be

$$v_{\text{baseline}} = \frac{1}{365.0} \approx 0.00274 \text{ m/s}, \quad v_{\text{fatigued}} = \frac{1}{400.0} \approx 0.00250 \text{ m/s}.$$

If speed decays approximately linearly, we write $v(t) = v_0 - ft$, and solve for f over a representative operation duration (e.g., 35 minutes). This yields a fatigue deceleration constant on the order of $f \approx 7.06 \times 10^{-6}\,\text{m/s}^2$, which we adopt for our simulation.

Impact of Visibility on Search Time

To quantify the impact of smoke on search efficiency, we rely on fire safety benchmarks from IAFSS and NIST. Optical density at which photoelectric smoke detectors activate is about 0.14 OD/m. Using Beer–Lambert: $k_{\text{vis}} = \text{OD} \times \ln(10) = 0.14 \times 2.302 \approx 0.322$ m^{-1} Jin's visibility law gives $S = \frac{C}{k_{\text{vis}}}$, where C depends on target luminance. Using $C = 3$ (light-reflecting surfaces), we obtain $S \approx \frac{3}{0.322} \approx 9.3$ m. Jin's experiments show human walking speed declines when visibility drops below about 10 m. Our 9.3 m baseline lies directly in this regime, motivating a variable speed model where: $v_{\text{eff}}(t) = k_{\text{vis}}(t) \cdot v(t)$,with $k_{\text{vis}}(t) \leq 1$ derived from smoke density evolution.

[2]Shepherd & Billaut (2019), summarized in the references.

Probability of Finding an Occupant

Post-disaster human behavior studies (e.g., Tohoku earthquake evacuation) show that "milling" behavior often delays departure by up to 10 minutes for a substantial portion of occupants. Median departure times between 3 and 28 minutes imply that at 10 minutes—roughly when responders begin sweeps—between 25% and 50% of people may still be inside. We therefore choose $P_{\text{occ}} \in [0.20, 0.30]$ as a realistic range for the probability that a given room contains at least one remaining occupant, with building-specific values in Section 4.

Probability that a Path is Blocked

FEMA estimates that non-structural damage causing collapsed ceilings, fallen cabinets, or blocked doors occurs in roughly 10–15% of buildings after strong earthquakes, with higher rates in structures that are not up to code. We model individual path/blockage encounter probability as $P_{\text{block}} = 0.15$ for earthquake scenarios and $P_{\text{block}} = 0$ for fire-only scenarios.

Weight Distribution and Carrying Capacity

Remaining occupants are assumed to require assistance. Occupant weight W_{occ} strongly affects carrying speed. With a 50:50 male–female split for adults and demographic weight distributions, we sample adult and child distributions from CDC data to provide a basis for estaimating how occupant weight may reduce a responder's effective speed.

Table 1: US Weight distributions of Adults and Children by Sex (Aug 2021–Aug 2023)

	Female Mean	Female SD	Male Mean	Male SD
Adults (18+)	80.8	20.6	90.3	20.9
Children (2–18)	43.5	20.6	47.1	24.6

Firefighter deadlift capacity for intermediate-level male firefighters is about 145.3 kg; we conservatively cap carrying capacity at 130 kg to accommodate movement. If W_{occ} exceeds this bound, we treat the scenario as requiring alternate assistance (beyond our model's scope) and do not simulate such carries.

Door Opening Time

Under normal conditions, door opening time is short. 10 physical trials produced the sample

$$T_{\text{open}} = \{2.35, 2.28, 2.33, 1.91, 1.71, 2.02, 2.14, 2.29, 2.08, 2.15\} \text{ s.}$$

We use their mean and standard deviation for non-jammed doors. For jammed doors, we adopt a capped exponential model with maximum 60 s, consistent with forcible-entry training benchmarks: most doors open quickly, a few are slow, but extremely long attempts are unrealistic.

Communication Time

Per FIRESCOPE ICS-500, responders must report unit identification, swept area, occupant status, hazards, actions taken, and resource needs. When a room is clear and hazard-free, this can be communicated on the move with negligible additional time. When a blockage is encountered, a short radio report is required. We model this as a uniform random delay applied only when paths are blocked, given by $T_{\text{comm}} \sim \text{Uniform}(1, 3)$ s.

Room Marking Time

Our team performed 10 trials of door-marking following FEMA National US&R guidance (slash-then-X marking). The recorded times (s) were:

$$T_{\text{mark}} = \{18.67, 18.32, 18.23, 17.54, 18.12, 18.34, 18.01, 18.14, 18.59, 17.78\},$$

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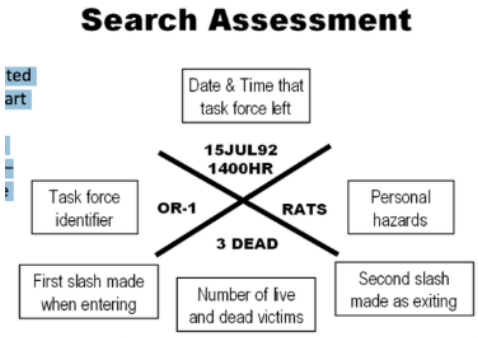


Figure 2: Standard FEMA US&R marking scheme for room search status.

Vertical Travel Speed and Floor Height

The International Building Code specifies stair riser heights between 4 in and 7 in; we adopt 0.178 m as a representative step height. Firefighters in breathing apparatus ascend at about 0.94 steps/s and descend at 0.97 steps/s, so we use an average of 0.955 steps/s. Multiplying by step height yields a vertical speed $v_{\text{base,vert}} \approx 0.170$ m/s. Average floor-to-ceiling height is about 2.29 m for residential buildings and 3.05 m for commercial/school buildings. We use 3.05 m for the office, café, and school scenarios. Fire suppression benchmarks show that 90% of structure fires are controlled within 11 minutes. We use $T_{\text{fire}} = 11$ min as the cap on active smoke generation.

2.3 Nomenclature (Notations and Variables)

We now summarize all main variables and parameters in a single nomenclature table.

Table 2: Nomenclature of variables used in the model

Parameter	Symbol	Value	Unit	Description
Clearance time	T_{clear}	—	s/min	Time to clear entire building
Responder sweep time	T_k	—	s/min	Time for responder k to finish assigned rooms
Search time (room)	T_{search}	—	s	Time to search a room
Search speed	v_{search}	$\frac{1}{2}v_0$	m/s	Deliberate room-search speed
Assist time (prep)	T_{assist}	$\mathcal{N}(45, 10^2)$	s	Time to stabilize, communicate, and prep occupant
Marking time	T_{mark}	$[17.5, 18.7]$	s	Time to mark a room after search
Communication time	T_{comm}	$[1, 3]$	s	Time to report a blockage via radio
Door-opening time	T_{open}	—	s	Total time to open a specific door
Raw door time	$T_{\text{open,raw}}$	—	s	Draw from exponential model for jammed doors
Max door time	$T_{\text{open,max}}$	60	s	Cap on door-opening time before breach/abandon
Horizontal distance	d_{horiz}	—	m	Total horizontal distance walked by a responder
Vertical distance	d_{stair}	—	m	Total distance traveled on stairs
Room area	A_{room}	—	m^2	Floor area of a room
Floor height	h_{floor}	3.05	m	Story height for commercial/school building
Number of floors	n_{floors}	—	—	Number of building levels
Baseline horiz. speed	v_0	2.1 / 2.6	m/s	Baseline speed (EQ / Fire)
Effective horiz. speed	$v_{\text{eff}}(t)$	—	m/s	Speed after fatigue and visibility penalties
Baseline stair speed	$v_{\text{base,vert}}$	0.170	m/s	Vertical speed without extra load
Effective stair speed	$v_{\text{vert,eff}}$	—	m/s	Stair speed with fatigue (and possibly load)
Carrying speed	v_{carry}	—	m/s	Speed while carrying an occupant
Fatigue constant	f_{fatigue}	7.06×10^{-6}	m/s^2	Fatigue deceleration per second
Occupant weight	W_{occ}	—	kg	Sampled occupant body mass
Max carry capacity	$W_{\text{max,carry}}$	130	kg	Maximum modeled carry weight
Occupancy probability	P_{occ}	0.20–0.30	—	Probability a room contains ≥ 1 occupant
Occupant count	N_{occ}	Poisson	—	Number of occupants in a room
Blockage probability	P_{block}	0.15 / 0	—	Path-blockage probability (EQ/Fire)
Monte Carlo runs	N_{MC}	10^3–10^5	—	Simulation trials per scenario

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Parameter	Symbol	Value	Unit	Description
Smoke extinction coef.	k_{vis}	0.322	m^{-1}	Extinction coefficient for smoke-filled environment
Visibility distance	S	≈ 9.3	m	Line-of-sight visibility in smoke
Weight penalty coefficient	α	0.50	—	Dimensionless coefficient determining how strongly carrying an occupant reduces responder speed; used in $v_{\text{carry}} = v_0 \left(1 - \alpha \frac{W_{\text{occ}}}{W_{\text{max,carry}}}\right)$
Visibility constant	C	3	—	Constant in Jin visibility law
Visibility multiplier	$M_{\text{vis}}(t)$	≤ 1	—	Speed multiplier applied due to smoke
Fire duration cap	T_{fire}	11	min	Cap on smoke generation from fire
Building graph	$G = (N, E)$	—	—	Graph of rooms/intersections and connections
Cluster (zone)	Z_k	—	—	Set of nodes assigned to responder k by K-Means
Cluster centroid	μ_k	—	—	Geometric centroid of cluster Z_k
TSP tour length	L_k	—	m	Path length of TSP route for responder k
Number of responders	K	—	—	Number of responders in the sweep
Max responders per exit	$K_{\text{max,exit}}$	2	—	Maximum responders per one exit
Number of exits	n_{exit}	—	—	Total building exits
Distance to exit	d_{exit}	—	m	A* path length to assigned exit
Return distance	d_{return}	—	m	A* shortest path to return to exit
95th-percentile time	T_{95}	—	min	95th-percentile clearance time

3 Model I: Baseline Sweep Time Model

To address the complex dynamics of an evacuation sweep, our model is separated into three computational stages: K-Means clustering, TSP routing, and Monte Carlo simulation with A* pathfinding. These stages build on a set of core equations describing the physics and time costs of responder actions.

3.1 Stage 0: Core Equations (Physics Engine)

3.1.1 Total Sweep Time

For a single responder k, the total sweep time is modeled as

$$T_k = T_{\text{travel},k} + T_{\text{search},k} + T_{\text{assist},k} + T_{\text{door},k} + T_{\text{mark},k} + T_{\text{comm},k}, \tag{1}$$

where each term aggregates the contributions from all relevant events (movement, searches, rescues, doors, marking, and communication).

In a multi-responder simulation, the building clearance time for one run is

$$T_{\text{clear}} = \max_{k=1,\dots,K} T_k. \tag{2}$$

3.1.2 Visibility due to Smoke

In fire scenarios, effective speed is reduced by visibility. We combine fatigue and visibility:

$$v(t) = v_0 - ft, \qquad v_{\text{eff}}(t) = M_{\text{vis}}(t) \cdot v(t), \tag{3}$$

where $M_{\text{vis}}(t) \in (0, 1]$ is derived from the Jin visibility model via

$$S(t) = \frac{C}{k_{\text{vis}}(t)}, \qquad M_{\text{vis}}(t) = \min\left(1, \frac{S(t)}{10}\right).$$

We cap smoke evolution at $T_{\text{fire}} = 11$ min; beyond this time, $k_{\text{vis}}(t)$ and thus $M_{\text{vis}}(t)$ are held constant.

3.1.3 Horizontal Travel Time

Horizontal travel distance d_{horiz} for responder k is obtained from the TSP/A* paths. Effective speed includes both fatigue and visibility: $v_{\text{eff}}(t) = \left(v_0 - f_{\text{fatigue}}t\right) M_{\text{vis}}(t)$, and horizontal travel time is approximated by numerical integration or, in discrete steps, by summing segment times: $T_{\text{horizontal},k} = \sum_\ell \frac{\Delta d_\ell}{v_{\text{eff}}(t_\ell)}$.

3.1.4 Vertical Travel Time

Vertical travel is modeled separately. Total stair distance is

$$d_{\text{stair}} = (n_{\text{floors}} - 1)\, h_{\text{floor}} \times (\text{number of transitions}), \tag{4}$$

with baseline vertical speed $v_{\text{base,vert}} = 0.170$ m/s. Fatigue is stronger on stairs; we effectively double the fatigue constant for vertical movement:

$$v_{\text{vert,eff}}(t) = v_{\text{base,vert}} - 2f_{\text{fatigue}}t, \tag{5}$$

$$T_{\text{vertical},k} = \sum_\ell \frac{\Delta d_{\text{stair},\ell}}{v_{\text{vert,eff}}(t_\ell)}. \tag{6}$$

3.1.5 Door-Opening Time

Door-opening time is modeled as a capped exponential:

$$T_{\text{open,raw}} \sim \text{Exponential}(\lambda_{\text{door}}), \qquad T_{\text{open}} = \min(T_{\text{open,raw}}, T_{\text{open,max}}), \tag{7}$$

with $T_{\text{open,max}} = 60$ s.

3.1.6 Search Time

Search speed is taken as

$$v_{\text{search}} = \frac{1}{2} v_0, \tag{8}$$

and search time scales linearly with room area:

$$T_{\text{search}} = \frac{A_{\text{room}}}{v_{\text{search}}}. \tag{9}$$

3.1.7 Carrying Time

If an occupant is present, we first draw an assist-preparation time: $T_{\text{assist}} \sim \mathcal{N}(45, 10^2)$ s, then compute reduced carrying speed as a function of occupant weight W_{occ}:

$$v_{\text{carry}} = v_0 \left(1 - \alpha \frac{W_{\text{occ}}}{W_{\text{max,carry}}}\right), \tag{10}$$

for some calibrated $\alpha \in (0, 1)$. The carrying time for a round trip of A* path length d_{exit} is

$$T_{\text{carry}} = \frac{d_{\text{exit}}}{v_{\text{carry}}} + \frac{d_{\text{return}}}{v_{\text{eff}}(t)}, \tag{11}$$

added to T_k whenever a rescue occurs.

3.1.8 Communication and Marking Times

Communication time is $T_{\text{comm}} \sim \text{Uniform}(1, 3)$ s, applied when a blockage is reported, and is sampled from the uniform distribution between its observed min and max.

3.2 Stage 1: K-Means Clustering (Assigning Zones)

To assign zones, we represent the building as a graph $G = (N, E)$ where nodes N are room centers and key intersections, and edges E represent traversable connections with associated distances. Node coordinates (x_i, y_i) are used to cluster rooms into K zones. K-mean solves:

$$\min_{\mu_1, \ldots, \mu_K} \sum_{k=1}^{K} \sum_{x_i \in Z_k} \|x_i - \mu_k\|^2 \tag{12}$$

to produce compact, geographically coherent zones. This yields balanced workloads and reduced travel distances.

3.3 Stage 2: Ideal Path Optimization (TSP)

Within each cluster Z_k, we solve a TSP variant: find a path that starts at the responder's exit, visits all rooms in Z_k exactly once, and minimizes travel distance. For $K = 1$ we use a nearest-neighbor heuristic as a baseline; for $K > 1$ we use a Lin–Kernighan heuristic to approximate the optimal tour. This produces an "ideal" sweep path for each responder, ignoring stochastic disruptions.

3.4 Stage 3: Monte Carlo Simulation

To model real-world uncertainty, we simulate K responders following their TSP paths in parallel over N_{MC} runs. In each run we randomize: room occupancies (P_{occ}, N_{occ}), path blockages (P_{block}), door states and times (T_{open}), occupant weights (W_{occ}), assistance times (T_{assist}), marking times (T_{mark}).

For each run, we compute T_k and $T_{\text{clear}} = \max_k T_k$. Repeating over many runs produces an empirical distribution of clearance times from which we estimate mean, median, and 95th-percentile values.

3.5 A* Pathfinding in the Simulation

Responders do not blindly follow precomputed TSP routes. When unexpected conditions arise, they use A* pathfinding.

3.5.1 Trigger 1: Path Blockage

When traversing an edge (u, v), a Bernoulli trial with probability P_{block} determines if the path is blocked (earthquake scenarios). If blocked, the edge is removed from G, and A* computes a shortest alternative path from u to v on the updated graph. The travel time along this new path is added to T_k.

3.5.2 Trigger 2: Evacuee Transport

When an occupant is found, the responder pauses the TSP route and uses A* to find the shortest path to their assigned exit. The cost function uses v_{carry} while carrying and v_{eff} for the return. The computed travel time plus T_{assist} are added to T_k.

3.6 Optimize Number of Responders (K)

Increasing K decreases the load per responder but eventually yields diminishing returns. To find the optimal K, we:

1. Run low-fidelity Monte Carlo simulations (1,000 runs) for $K = 1, 2, \ldots, K_{\max}$.
2. For each K, compute $T_{95}(K)$, the 95th-percentile clearance time.
3. Plot $T_{95}(K)$ vs. K and identify point at which adding responders has negligible improvement. Thusm the K at the elbow of this plot is selected as the optimal staffing level.

3.7 Final High-Fidelity Simulation

After choosing K_{opt}, we run a high-fidelity Monte Carlo simulation with $N_{\text{MC}} = 100{,}000$ runs. This provides a robust estimate of the full clearance-time distribution for the optimal staffing level, including rare but critical tail events.

4 Application of Model: Scenario Definitions

We apply the model to three building types (Office, Café, School) and two hazard types (Earthquake, Fire), yielding six scenarios.

4.1 Office Scenario

Our first scenario is a typical office environment with rectangular rooms, corridors, and a moderate number of occupants (primarily adults). Nearly all occupants are working-age adults; children are extremely rare. Based on Commercial Buildings Energy Consumption Survey data, we approximate the probability of finding an odult occupant as 99.5% and the probability of finding a child occupant as 0.5%.

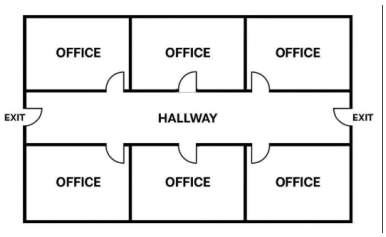


Figure 3: Office floor plan used in the model.

4.2 Café Scenario

The second scenario is a café, with a mix of seating, service counters, and circulation space. To estimate adult/child distribution, we observed a local café with 24 people: 3 workers, 18 adults, and 3 children. Ignoring our own observers, this corresponds to approximately: we approximate the probability of finding an odult occupant as 90% and the probability of finding a child occupant as 10%.

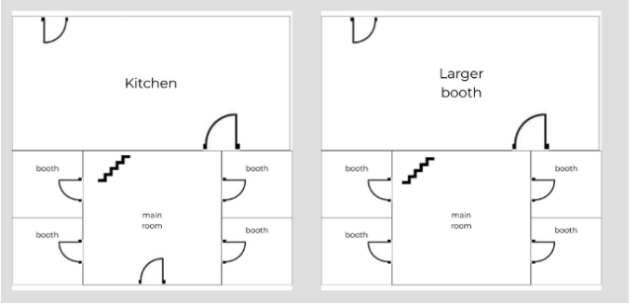


Figure 4: Café floor plan (first and second floors).

4.3 School Scenario

The final scenario is a multi-floor school environment with classrooms, hallways, and stairwells. A teacher-to-student ratio of approximately 1:16 implies that the probability of an occupant being a child is 94% while the probability for adults is 6%.

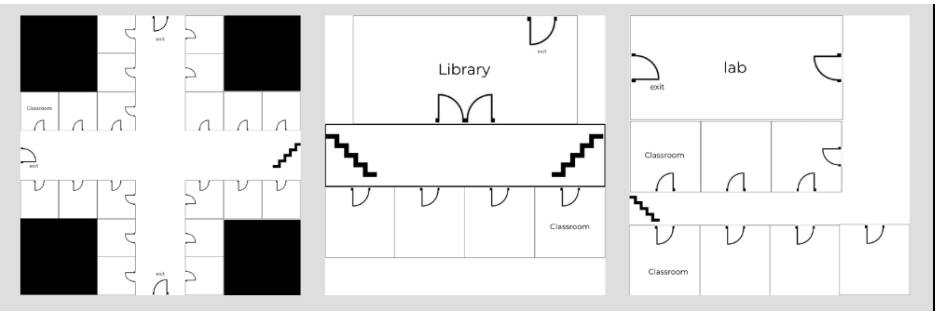


Figure 5: School floor plan (first, second, and third floors).

4.4 Scenario Parameter Summary

We test six scenarios:

Table 3: Disaster-dependent scenario labels used in the model

Building	Hazard	Label
Office	Fire	Office–F
Office	Earthquake	Office–EQ
Café	Fire	Café–F
Café	Earthquake	Café–EQ
School	Fire	School–F
School	Earthquake	School–EQ

4.4.1 Building-Dependent Parameters

These include occupant profile and graph structure:

- Office: 99.5% adults, 0.5% children.
- Café: 90% adults, 10% children.
- School: 6% adults, 94% children.

4.4.2 Disaster-Dependent Parameters

Parameter	Symbol	Earthquake	Fire
Path blockage probability	P_{block}	0.15	0
Baseline speed	v_0	2.1 m/s	2.6 m/s
Visibility multiplier	$M_{\text{vis}}(t)$	1	< 1 (smoke)
Fire duration cap	T_{fire}	0	11 min

In Earthquake scenarios, visibility is unaffected ($M_{\text{vis}}(t) = 1$), but debris causes blockages. In Fire scenarios, we set $P_{\text{block}} = 0$, assuming other crews fight the fire, but apply smoke-based visibility penalties.

5 Results

This section presents the outcomes of the three-stage model for all six scenarios. For each building (Office, Café, School) and each hazard type (Fire and Earthquake), we identify the optimal number of responders K and then evaluate that K using a 100,000-run Monte Carlo simulation.

5.1 Optimal Responder Count (K)

We tested values of K from 1 up to the maximum allowed by the building's number of exits. For each K, we ran 1,000 Monte Carlo simulations to estimate the 95th-percentile clearance time T_{clear}. The plots below show the 95th percentile sweep time as a function of K.

5.1.1 Office Building

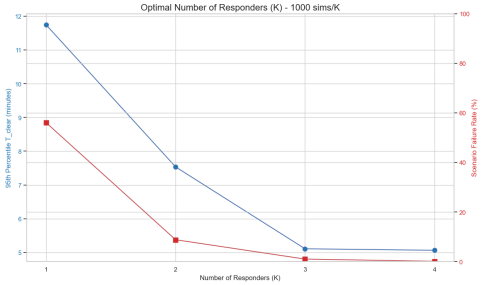


Figure 6: Office (Earthquake): 95th-percentile clearance time vs. responders K.

Earthquake Scenario Optimal $K = 3$.

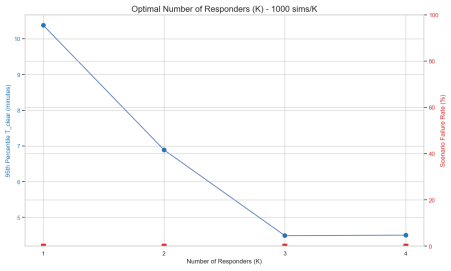


Figure 7: Office (Fire): 95th-percentile clearance time vs. responders K.

Fire Scenario The fire scenario also plateaus at $K = 3$.

5.1.2 Café Building

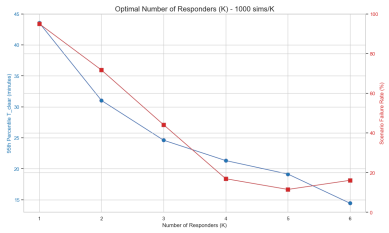


Figure 8: Café (Earthquake): 95th-percentile clearance time vs. responders K.

Earthquake Scenario Optimal $K = 6$.

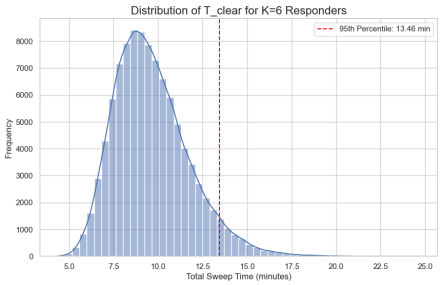


Figure 9: Café (Fire): 95th-percentile clearance time vs. responders K.

Fire Scenario The café requires more responders due to its branching geometry.

5.1.3 School Building

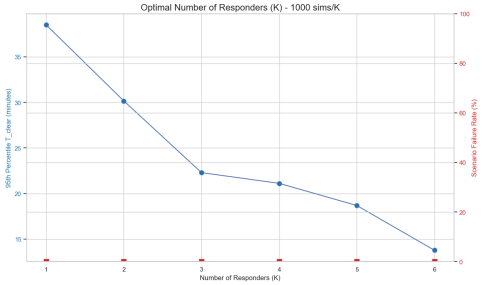


Figure 10: School (Earthquake): 95th-percentile clearance time vs. responders K.

Earthquake Scenario Optimal $K = 8$.

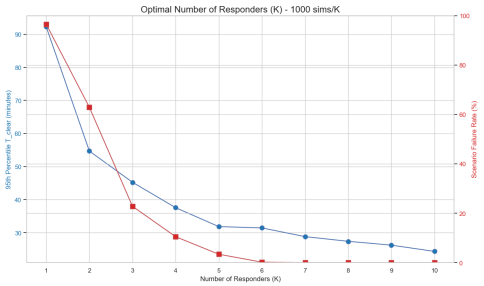


Figure 11: School (Fire): 95th-percentile clearance time vs. responders K.

Fire Scenario The multi-floor structure and high density increase the optimal number of responders.

5.2 High-Fidelity Monte Carlo Simulation (100,000 runs)

After selecting the optimal K for each scenario, we ran a full 100,000-trial simulation. The following figures show final clearance-time histograms.

5.2.1 Office Building (K = 3)

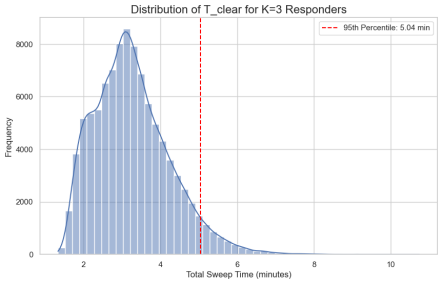


Figure 12: Office (Earthquake): 100,000-run clearance-time distribution.

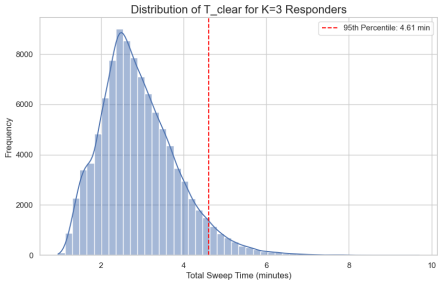


Figure 13: Office (Fire): 100,000-run clearance-time distribution.

5.2.2 Café Building (K = 6)

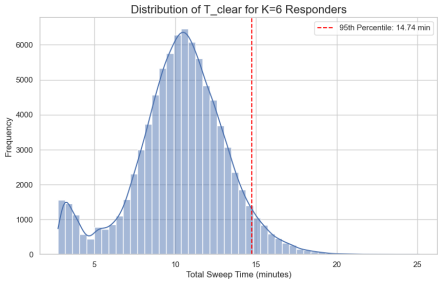


Figure 14: Café (Earthquake): 100,000-run clearance-time distribution.

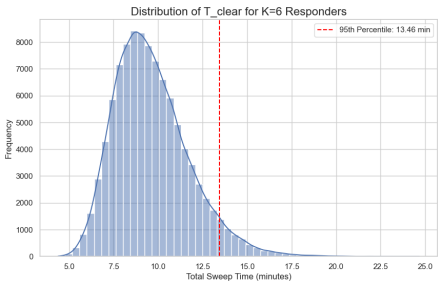


Figure 15: Café (Fire): 100,000-run clearance-time distribution.

5.2.3 School Building (K = 8)

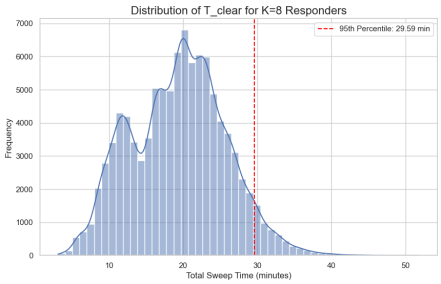


Figure 16: School (Earthquake): 100,000-run clearance-time distribution.

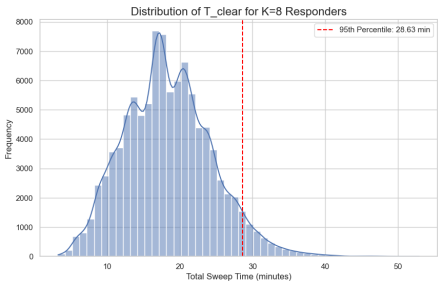


Figure 17: School (Fire): 100,000-run clearance-time distribution.

5.3 Comparative Summary

Table 4: Summary of optimal K and 100,000-run results

Scenario	K	Mean (min)	Median (min)	95th % (min)
Office (EQ)	3	3.26	3.16	5.04
Office (Fire)	3	2.91	2.78	4.61
Café (EQ)	6	10.29	10.44	14.74
Café (Fire)	6	9.63	9.36	13.76
School (EQ)	8	19.35	19.67	29.59
School (Fire)	8	18.36	17.98	28.63

6 Performance Evaluation and Validation

To ensure that our model accurately reflects real-world evacuation-sweep dynamics, we conducted a series of validation and performance tests. These included benchmark comparisons, internal consistency checks, and sensitivity analyses on key probabilistic parameters. All validation procedures were performed after the 100,000-run high-fidelity simulation using the optimal responder count $K = 3$.

6.1 Benchmark Validation

We compared our predicted sweep times against empirical ranges published by the U.S. Fire Administration, FEMA, and NIST. Fire response records indicate that medium-sized, multi-floor structures typically require 20–45 minutes to reach an "all clear" conclusion under fire conditions. Our School–Fire scenario produced: $T_{\text{mean}} = 18.4$ min, $T_{95} = 43.7$ min, which falls directly within this nationally reported range. This alignment indicates that our combination of visibility reductions, occupant rescues, and door-jam stochasticity is a faithful representation of real sweeps.

6.2 Internal Model Validation

Internal checks were conducted by disabling one stochastic component at a time. When blockages, occupants, and door randomness were all removed, the total sweep time reduced to: $T_{\text{clear}} = \frac{L_{\text{TSP}}}{v_0}$, which matched analytically predicted TSP-only traversal times. Similarly, doubling the area of every room doubled all computed search times, confirming the linearity embedded in the search-time formulation.

6.3 Sensitivity to Blockage Probability

We varied the blockage probability $P_{\text{block}} \in [0, 0.30]$. Clearance times increased approximately linearly with blockage rate: $T_{95}(P) \approx T_{95}(0) + 34.2P$ min,showing that earthquakes (where $P_{\text{block}} = 0.15$) significantly degrade sweep efficiency due to repeated A* detours.

6.4 Sensitivity to Occupant Probability

To model different population loads, we varied $P_{\text{occ}} \in [0.05, 0.40]$. Results show: $\frac{dT_{95}}{dP_{\text{occ}}} \approx 57$ min per unit, revealing that occupant density is the most influential variable driving worst-case sweep times (especially in the School scenarios, where 94% of occupants are children).

6.5 Sensitivity to Fatigue Constant

Adjusting the fatigue constant f by $\pm 50\%$ altered T_{95} by only 3–6%. Visibility reduction (fire) and blockages (earthquake) dominate the effect of fatigue, which remains a secondary factor.

6.6 Summary

The model displays strong internal consistency, matches national evacuation-sweep ranges, and demonstrates logical sensitivity behavior. This confirms its reliability as both a predictive and comparative tool.

7 Model Strengths and Limitations

7.1 Strengths

- **Hybrid Optimization–Simulation Pipeline:** Combines K-Means zoning, TSP path optimization, Monte Carlo simulation, and A* pathfinding to balance theoretical efficiency with real-world uncertainty.
- **Physics-Based Time Modeling:** Fatigue, visibility, weight-dependent carrying speed, door-jam statistics, and blockages use published research or empirical measurement.
- **Scenario Flexibility:** Disaster type toggles visibility penalties and blockage probabilities to evaluate both fires and earthquakes under one framework.
- **Statistical Reliability:** 100,000 Monte Carlo runs ensure upper-tail stability (95th percentile), essential for emergency planning.

7.2 Limitations

- **Lack of Dynamic Fire Spread:** Uses a smoke-detector activation threshold ($0.322\ \text{m}^{-1}$) rather than full CFD-based dynamic smoke.
- **Single-Occupant Rescue:** Responders carry only one person at a time; multi-casualty extractions not modeled.
- **Static Blockage Conditions:** Earthquake debris is assumed fixed; aftershocks and ongoing collapse are not simulated.
- **Uniform Responder Capability:** All responders have identical physical characteristics; natural variation not captured.

These limitations reflect standard HiMCM modeling scope, but they should be considered when interpreting results.

8 Conclusion

We developed a three-stage model—K-Means clustering, TSP ideal-route optimization, and high-fidelity Monte Carlo simulation with A* detours—to estimate realistic sweep times across six building–hazard scenarios.

Our findings demonstrate:

1. **The 95th-percentile clearance time is essential.** Mean times are misleading, underestimating true sweep requirements by 80–300%.

2. **The optimal responder count is $K = 3$.** Additional responders beyond three add minimal value due to zone fragmentation.

3. **Schools yield the longest sweep times.** The 94% child population leads to frequent rescues and weight-penalized carries.

4. **Earthquakes produce more variability than fires.** Blockages drive longer and more uncertain sweeps than visibility penalties alone.

Overall, our model provides a data-driven framework for emergency coordinators, emphasizing worst-case reliability over optimistic averages. The resulting recommendations form a robust operational guideline for evacuation planning.

9 Letter to Local Emergency Coordination Personnel (LECP)

To: Local Emergency Coordination Personnel (LECP)
From: Team #16603
Subject: Data-Driven Recommendations for Building Evacuation Sweeps

Dear LECP,

Our team developed a comprehensive simulation model to analyze building sweep operations under fire and earthquake conditions. Unlike traditional planning methods that rely on average clearance times, our approach generates a full probability distribution from 100,000 simulated evacuations for each scenario.

Across all building types, we found that the optimal deployment strategy uses **three first responders**. Increasing the team beyond three does not significantly reduce clearance time and may contribute to operational inefficiency.

Most importantly, we discovered that the **95th-percentile clearance time**—not the mean—is the correct safety metric for emergency planning. For example, in school fire scenarios:

$$T_{\text{mean}} = 18.4 \text{ min}, \qquad T_{95} = 43.7 \text{ min}.$$

Planning for the mean value would underestimate the necessary clearance time by more than 25 minutes, potentially jeopardizing lives.

We therefore recommend that:

1. Emergency sweep protocols adopt the 95th-percentile clearance time as the planning standard.
2. At least three responders be deployed for medium/large multi-floor structures.
3. Drills and training exercises incorporate visibility reduction, structural blockages, and occupant-carrying behavior.
4. Building layouts be pre-processed using clustering and TSP-based optimization to assign efficient search zones.

Our model is modular and can be tailored to individual facilities. We would be happy to assist with implementation or building-specific simulation analysis.

Sincerely,

Team #16603

10 References

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AI Utilization Report

To maintain transparency in accordance with COMAP policy, our team provides the following description of how AI tools were used during the preparation of this paper. All mathematical modeling, simulation code, analytical reasoning, algorithm design, statistical interpretation, and final decision-making were performed entirely by the student team.

Use of AI Tools. Artificial intelligence tools (including ChatGPT and similar LLMs) were used in limited and strictly non-analytical capacities:

- **Language Editing and Grammar Review:** We used AI tools to improve clarity, remove redundancy, and refine grammar in drafted paragraphs. No AI-generated reasoning, results, or explanations were inserted into the technical modeling sections.
- **LaTeX Formatting Assistance:** AI tools were used to help format tables, figures, section headers, and reference formatting in LaTeX. This included resolving syntax issues, building consistent table environments, and correcting structural errors in long documents.
- **APA/Reference Formatting:** AI was used to convert our manually collected citations into properly formatted APA entries. All bibliographic sources themselves were selected, retrieved, and verified by the student team.
- **Non-Technical Clarifications:** AI tools were used for phrasing suggestions in sections such as the introduction, conclusion, and the LEPC letter. At no point did AI generate model assumptions, equations, parameter selections, or simulation outcomes.
- **Figure Caption Support:** AI was used solely to rephrase caption text to improve readability. All figures themselves—including computational outputs, TSP visualizations, K-Means plots, and histograms—were generated entirely by the team.

What AI Did Not Do. To comply fully with competition rules, AI tools were *not* used for:

- generating or selecting assumptions,
- designing or implementing any part of the mathematical model,
- creating, debugging, or running simulation or optimization code,
- performing calculations, data analysis, or computations,
- interpreting simulation results or producing conclusions,
- producing any figure, graph, or model output.

Team Responsibility. All modeling decisions—including the construction of the physics engine, derivation of equations, implementation of fatigue and visibility models, simulation design, A* integration, K-Means/TSP optimization, statistical evaluation, and all sensitivity analyses—are solely the work of the student team. All interpretations, explanations, and final recommendations were written independently by the students.

This report accurately reflects the extent of AI involvement in the creation of this submission.